

Open-path dual-comb spectroscopy to an airborne retroreflector

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We demonstrate a new technique for spatial mapping of multiple atmospheric gas species. This system is based on high-precision dual-comb spectroscopy to a retroreflector mounted on a flying multicopter. We measure the atmospheric absorption over long open-air paths to the multicopter with comb-tooth resolution over 1.57–1.66 μm , covering absorption bands of CO_2 , CH_4 , H_2O , and isotopologues. When combined with GPS-based path length measurements, a fit of the absorption spectra retrieves the dry mixing ratios versus position. Under well-mixed atmospheric conditions, retrievals from both horizontal and vertical paths show stable mixing ratios as expected. This approach can support future boundary layer studies as well as plume detection and source location. © 2017 Optical Society of America

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Open-path dual-frequency-comb spectroscopy (DCS) has recently been shown to provide highly accurate and resolved broadband atmospheric spectra across kilometer-scale paths to fixed reflectors [1–3]. This makes DCS well suited to measurements and quantification of emissions of pollutants, hazardous gases, and greenhouse gases. In a recent direct intercomparison, two open-path DCS instruments retrieved atmospheric trace-gas mixing ratios to within 0.14%–0.4% over weeks, well below the natural atmospheric variability [3]. Here we describe an advance in open-path DCS that enables measurements of the horizontal and vertical spatial profile of atmospheric gases in near real time. This technique, especially when extended to other spectral regions, has a variety of potential applications, such as emissions quantification, boundary-layer profiling, and hazardous plume detection. For example, emissions from oil and gas facilities are important both for understanding global methane sources [4–6] and for understanding the impacts on ozone and aerosol formation [7–9]. By flying box-type patterns around a distributed source and using a mass-balance approach [10,11], such a system could be used to quantify CH_4 as well as volatile organic compounds. In addition, the mixing of gases within the planetary boundary layer is currently a major source of uncertainty in atmospheric transport models [12–14], which results in errors in source quantification using point sensors or satellites. Finally, this

technique could be used to rapidly scan an area for the presence of hazardous gases [15] and threat chemicals [16].

Unlike differential absorption LIDAR [17,18], which measures the atmospheric absorption at only a few laser frequencies, DCS measures the spectrum at many tens of thousands of individual frequencies with eye-safe near-infrared (or, in the future, mid-infrared) laser light [19,20]. As a consequence of this massively broadband nature, open-path DCS requires a reflector at the far end of the path to provide sufficient signal-to-noise ratio. Here, we demonstrate DCS to a retroreflector that is mounted on a multicopter, part of a small unmanned aircraft system (sUAS), as illustrated in Fig. 1. This combined sUAS-DCS system is used to scan horizontal and vertical paths and retrieve the column-integrated mixing ratios of water, CO_2 , and CH_4 . With future extensions across even broader near-infrared bandwidths and into the mid-infrared, such a system could provide spatial mapping across multiple gas species at high precision and accuracy.

The DCS system is described in detail in Ref. [21]. It is based on two near-infrared, self-referenced, optically coherent fiber frequency combs [22] with repetition rates of $f_r \sim 200$ MHz that differ by $\Delta f_r = 624$ Hz. A portion of each frequency comb is amplified, spectrally broadened in highly nonlinear fiber, combined in fiber, and filtered to cover 1.57–1.66 μm (or 6000–6330 cm^{-1}), which allows simultaneous measurements

of CO₂, CH₄, H₂O, and isotopologues. The mutual comb line-width was below 1 Hz and the absolute long-term linewidth was below 120 kHz, enabling ultra-high resolution sampling of the atmospheric spectrum with a point spacing of $f_r \sim 200$ MHz over $\sim 50,000$ comb teeth. The entire system is compact and portable [21], and has been operated in nonlaboratory environments [3,23].

The telescope system, sketched in Fig. 1(a), was designed for a large collection aperture, while remaining relatively lightweight and compact for rapid scanning. About 5 mW of the filtered light is sent to the launch telescope system for the open-path measurements. This launched power is below 9.6 mW at the telescope aperture, which is the Class 1 American National Standards Institute (ANSI) accessible emission limit (AEL) [24]. Only reflective optics were used to enable broad spectral bandwidth operation. The telescope uses two off-axis parabolic mirrors with a 3"-diameter collection aperture, resulting in a launched beam diameter of ~ 45 mm with a nearly diffraction-limited beam divergence of ~ 30 μ rad (half-angle). This telescope was mounted on a gimbal to allow pointing and tracking to the retroreflector. The gimbal was capable of both elevation and azimuthal motion with a velocity of 80 deg/s and an acceleration of 400 deg/s².

The multicopter (xFold Cinema x8 [25]) with payload is shown flying in Figs. 1(b) and 1(c). The primary payload is a commercially available, lightweight, 2.5"-diameter, 5-arcsec angular-deviation hollow corner-cube retroreflector. The use of a corner-cube retroreflector on the multicopter, rather than a plane mirror, drastically reduces the multicopter pointing requirements. Since the retroreflector has a wide field of view (approximately $\pm 15^\circ$), we can rely solely on the yaw control of the multicopter to maintain sufficient pointing back to the telescope. The payload included a radiosonde (i-Met RSB 1 [25]) for temperature, pressure, relative humidity, and real-time coarse GPS location

measurement. Finally, the payload also included a real-time kinematic GPS (RTK GPS, Swift Navigation Piksi [25]) for high-precision differential path length measurements.

The precision of the mixing ratios retrieved by the DCS depends on both received power and averaging time. Received powers range from a minimum of ~ 15 μ W up to typical levels of a few hundred microwatts, above which the detector saturates. At typical average received power levels of ~ 50 – 100 μ W, the measurement precision in ~ 10 s was 2 parts per million (ppm) of CO₂ and 16 parts per billion (ppb) CH₄, and improved to 0.6 ppm of CO₂ and 6 ppb of CH₄ in ~ 100 s (see [3] for a full Allan deviation). To support these averaging times, we used a "step scan" approach: the sUAS moves the retroreflector to a specific location where it hovers for a user-selected amount of time before moving to a new location. The total flight time for a single battery charge was ~ 15 min under our flight conditions.

The telescope system must track the motion of the retroreflector as it moves and hovers in order to obtain sufficient return power of the DCS comb light. Ideally, this tracking is within the ~ 30 - μ rad beam divergence, but this requirement is reduced (at the cost of reduced return power) by about a factor of two because of turbulence-induced fast angular jitter [26]. The telescope tracking was based on feedback from a focal plane array (FPA) that imaged the return from the retroreflector. Direct use of the near-infrared DCS light would require an expensive InGaAs camera and would not provide a large angular capture range. To circumvent these limitations, we launch a high-power (~ 1 W), low-divergence, 850-nm LED co-aligned with the DCS light. At this wavelength, the reflected LED light can be detected by a silicon FPA. The return LED light is bandpass filtered and imaged onto this FPA with a 500-mm-focal-length, 85-mm-diameter camera lens [typical image shown in Fig. 1(b)]. The images are read out at 30 Hz into a computer, averaged over two frames, smoothed by a 3-pixel Gaussian filter, and then processed to find the retroreflector location via two-dimensional peak finding. The offset of the identified intensity maximum from the target location [shown as crosshairs in Fig. 1(b)] was input to a proportional-plus-double-integral loop whose output controlled the telescope pointing at 15 Hz with a ~ 2 -Hz integration bandwidth, which was sufficient for these measurements. In the future, the feedback bandwidth could be improved by a higher frame rate camera and the tracking during movement could be improved with more sophisticated estimation algorithms (such as a Kalman filter). The retroreflected dual-comb light is collected by the off-axis telescope and focused onto a ~ 100 -MHz-bandwidth amplified InGaAs photodetector using a nonpolarizing beamsplitter, as shown in Fig. 1(a). The total loss of the telescope system (collected return power compared to input power to the telescope) was typically ~ 8 – 20 dB depending on atmospheric conditions (e.g., turbulence-induced scintillation or wind gusts impacting sUAS yaw stability). We achieved ~ 5 -dB higher overall collected light by use of a polarizing beamsplitter and quarter-wave plate to act as an optical circulator; however, this created additional etalons which negated the benefits of higher power.

For the analysis, we need the path length, air temperature, and air pressure. The path length is determined using either the radiosonde GPS and known GPS location of the telescope system, or preferably with the RTK GPS, which measures the differential path length between a receiver on the sUAS and a receiver located near the telescope system. The RTK GPS provides the path length

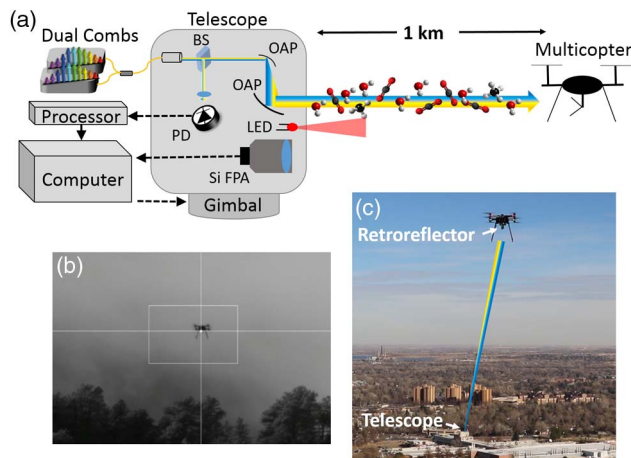


Fig. 1. Setup for DCS to a sUAS. (a) The light from both combs is combined in fiber, then launched from a telescope to a retroreflector on a small multicopter. A co-aligned 850-nm LED and silicon focal plane array (Si FPA) camera are used in the tracking servo to control the azimuth-elevation gimbal. PD, photodetector; BS, beam splitter; OAP, off-axis parabolic mirror. (b) Image from Si FPA showing the multicopter above the tree line. The bright spot is the retroreflected 850-nm LED light. A software feedback loop to the telescope gimbal centers this spot on the crosshairs, which simultaneously maximizes the DCS signal on the photodetector. (c) Photo of the small commercial multicopter. The dual-comb light is launched from the telescope located in the 1-km distant rooftop laboratory, as indicated.

to <6-cm accuracy at a 10-Hz sampling rate, but requires signals from at least seven GPS satellites and was available for only one of the two flight tests described below. We used an average temperature determined from the radiosonde on the sUAS and a sensor located near the telescope. The pressure is taken from the radiosonde.

The DCS signal is a series of interferograms that repeat at a rate of $\Delta f_r = 624$ Hz, or once every 1.6 ms. We coherently sum 100 of these interferograms on a field-programmable gate array and then further carrier-phase correct and sum N sets in software, yielding one interferogram (and thus spectrum) every $100 N / \Delta f_r$ seconds (here N ranged from 60 to 200, resulting in one spectrum every 10–30 s). Each spectrum is fit to an absorption model based on the high-resolution transmission (HITRAN) 2008 database [27] plus a piecewise-polynomial baseline to account for the comb spectral structure, as discussed in Refs. [1,3].

We flew two different flight patterns on December 1, 2016. The first flight pattern, shown by the yellow track in Fig. 2(a), consisted primarily of horizontal movement with respect to the telescope. Near real-time results are shown as a movie in Visualization 1. In Visualization 1, the lower right quadrant shows the real-time updates of the measured atmospheric spectrum and the lower left quadrant shows the real-time mixing ratios retrieved under a simplified assumption of a fixed 2-km path length, fixed temperature, and fixed pressure. In post-processing, we refit the raw spectra with the measured time-dependent path length, temperature, and pressure. Figure 2(b) shows the retrieved path-length-corrected mixing ratios as a function of time at 9.6-s averaging times along with the multicopter location and radiosonde GPS-measured path length. Prior to 13:07 local time, the multicopter was stationary on a platform. After takeoff at ~13:07, it hovered for 1–3 min at five different locations, as indicated in the position graphs. During transit the return power fluctuated, but it was generally well above the 15- μ W minimum threshold during hovering. From 13:16 to 13:17, the sUAS did not maintain sufficient multicopter yaw to point the retroreflector back to the telescope and signal was lost. Signal was regained at ~13:17, but subsequent loss of sUAS telemetry caused the multicopter to then return to the platform. The mixing ratios are fairly constant over the flight pattern as expected over this $\sim 100 \times 60$ m² area and prevailing ~ 1 m/s wind speed.

The second flight pattern consisted of a series of slant-column measurements—as illustrated in Fig. 3—from near ground level up to the 120-m (400-foot) ceiling height imposed by current Federal Aviation Agency (FAA) regulations [28]. Higher flights are possible with FAA approval and will be pursued in the future to study boundary layer variations of the trace gas concentrations. As with the horizontal path measurements, a move-hover pattern was used; the multicopter was flown to a specific altitude, where it hovered for ~ 4 min before ascending/descending to another altitude. From 14:33 to 14:36, the multicopter returned to the platform to exchange batteries. The overall flight duration was ~ 25 min. The RTK GPS was operational during the full flight. In addition, a fixed horizontal path was acquired simultaneously using a second output from the same DCS instrument directed by a second telescope to a fixed retroreflector located near the launch point of the multicopter.

The fitted mixing ratios for both paths, as well as corresponding sUAS height from the RTK GPS, are shown in Fig. 3(b). These data were acquired at wind speeds of ~ 0.5 –5 m/s and

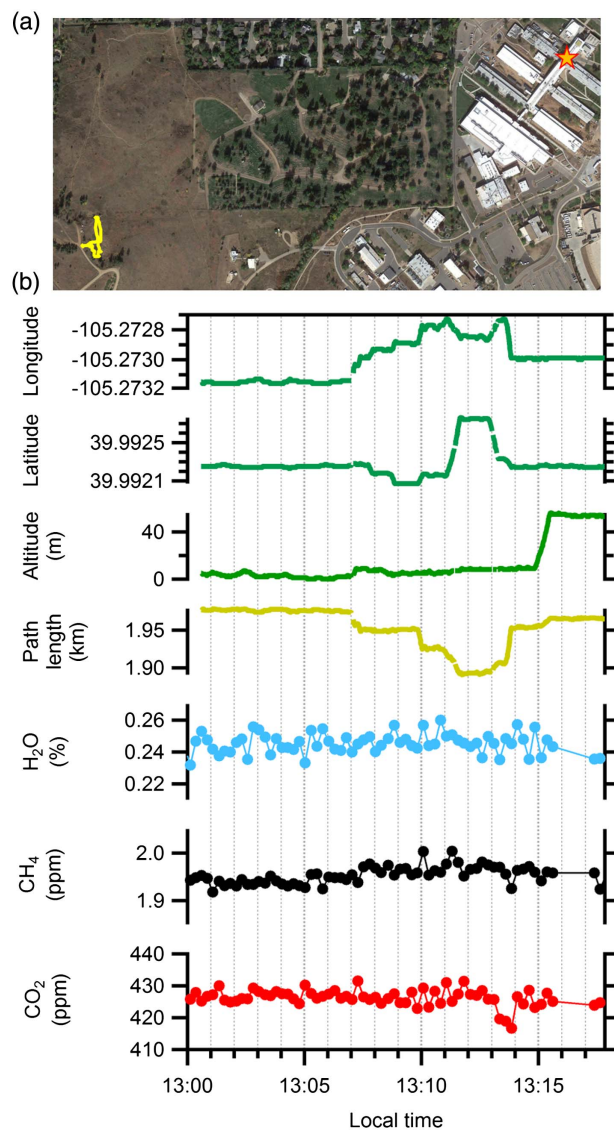


Fig. 2. (a) Flight path (yellow) recorded by the radiosonde GPS for the first flight pattern. The star indicates the DCS instrument and telescope location. Map data: Google, DigitalGlobe, U.S. Geological Survey, USDA Farm Service Agency. A video of the real-time data collection software for this flight is given in Visualization 1. (b) Mixing ratios for H₂O (blue), CH₄ (black), and CO₂ (red) obtained from the horizontal flight path shown in Fig. 2(b). These data are from spectra acquired every 9.6 s and are corrected for measured path length, temperature, and pressure. The upper four panels show the longitude, latitude, altitude, and path length derived from the radiosonde GPS.

with a forecast boundary layer height of ~ 2 km [29]; therefore, we also expect strong mixing of the atmosphere over the measured altitudes and thus relatively constant mixing ratios. We can determine the variation in slant column versus height by averaging all measurements at a given height bin, as is shown in Fig. 3(c). The residual scatter of these binned data is ~ 0.87 ppm CO₂, 5.7 ppb CH₄, and 0.0021% H₂O. As expected from the atmospheric conditions, there is no clear structure versus height. This lack of variability contrasts with the measurements of Ref. [3], acquired over a longer timescale and sometimes at lower wind speed, where significant temporal variations are observed both on diurnal timescales and shorter timescales as plumes traverse

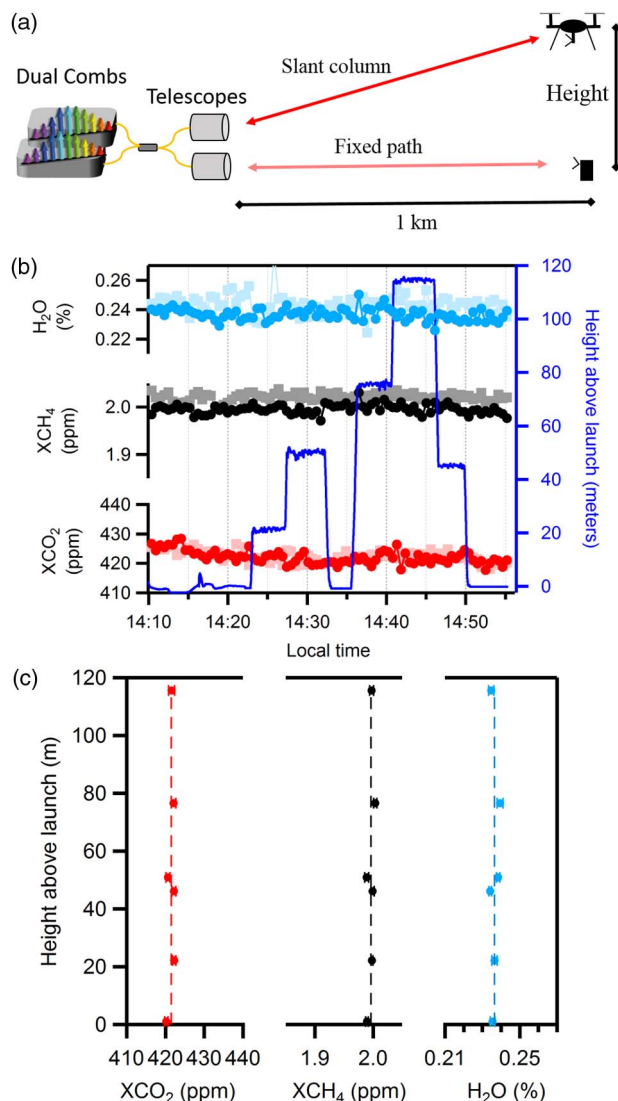


Fig. 3. Results of a flight to obtain a vertical concentration profile. (a) Diagram showing simultaneous slant column and fixed path measurements. (b) The path-corrected H_2O (medium blue), dry CH_4 (XCH_4 , black), and dry CO_2 (XCO_2 , red) mixing ratios obtained for 32-s-averaged data from the sUAS-DCS. The data in lighter colors were obtained along the fixed path. The corresponding retroreflector height above ground (from the RTK GPS) is also shown (right axis, dark blue). (c) Slant-column mixing ratio versus retroreflector height above ground from averaging approximately ten points over 4–5 min at each height in (b). The error bars are the standard error of the mean. Averaged over height (dashed line), the mixing ratios are 421.5 ± 0.9 ppm, 1.996 ± 0.006 ppm, and $0.236 \pm 0.002\%$ for CO_2 , CH_4 , and H_2O , respectively. The scatter about the mean is attributed to temporal, rather than spatial, variations under these well-mixed conditions.

the open-path beam. We do observe some temporal variability here, as seen for example in the ~ 5 -ppm decrease in XCO_2 over the measurement time in Fig. 3(b). However, this same XCO_2 decrease is observed simultaneously on the slant paths and fixed horizontal path, indicating that it reflects a temporal rather than vertical variability, again reflecting the well-mixed conditions. (The ~ 25 -ppb constant offset in CH_4 between the horizontal and slant path measurements is attributed to non-linearities when combining the two combs in fiber and has been

resolved [3].) In contrast, in less-well-mixed conditions or if the retroreflector height had exceeded the boundary layer, one would expect stronger variations. For example, photosynthesis can cause CO_2 gradients in the boundary layer of 1–10 ppm [30]. In regions with higher emissions, the variations in CO_2 or CH_4 between the boundary layer and free troposphere could be 20 ppm or more for CO_2 [31] and 100 ppb or more for CH_4 [32]. These changes are easily within our precision of <1 ppm for CO_2 and <6 ppb for CH_4 ; we intend to explore such vertical structures in future campaigns.

Here, we demonstrated simultaneous detection of CO_2 , CH_4 , and H_2O along a 2-km-roundtrip path using dual-comb spectroscopy with a retroreflector located on a sUAS. In the future, additional species could be detected with broader wavelength coverage and extension into the mid-infrared. Longer flight times will be available with the continued strong development of UAS. Finally, we have already demonstrated much longer path lengths of up to 12 km round-trip [2] with similar launched optical power levels. Operation at the ANSI maximum permissible exposure level of 100 mW/cm^2 [24] would allow for still longer path lengths. As sUAS flight times improve and FAA regulations evolve, more applications will become possible.

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